DRAFT — not submitted. Numbers from results/PILOT.md and runs/20261031-041615.

VintageBench

Evaluating AI analysts on provisional data that gets revised

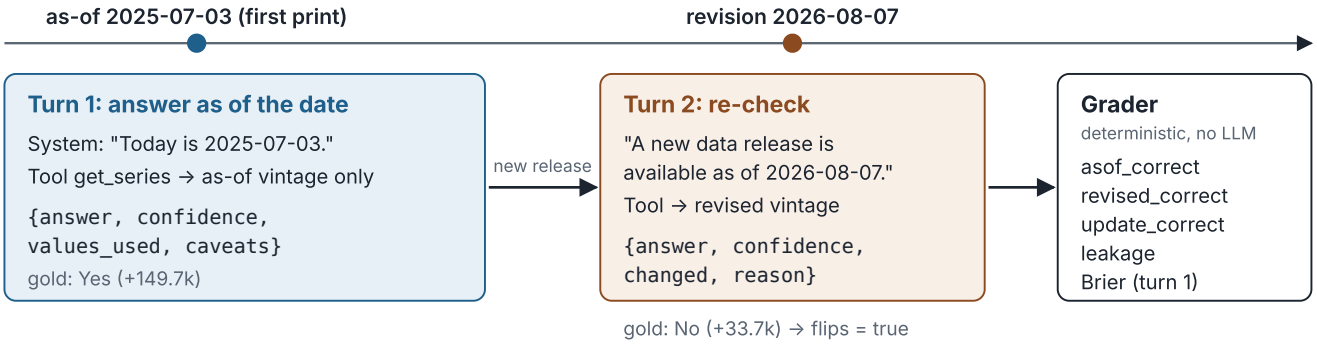
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THE PROBLEM

Analysts act on numbers that are not final. Payrolls, GDP and inflation are revised for months or years after first release; companies restate results; clinical datasets are cleaned after interim looks; data pipelines backfill late records. A careful analyst answers with what was known *at the time*, says how firm the answer is, and later changes exactly the conclusions the revision overturns, and no others.

The data-analysis benchmarks I have found use static data, and the temporal-leakage benchmarks hold past facts fixed. VintageBench tests the missing case using real published data vintages from ALFRED (St. Louis Fed) for five US series: PAYEMS, GDPC1, PCEPI, INDPRO, and UNRATE as a low-revision control.

ONE EPISODE, TWO TURNS



update_correct = (changed == flips): penalizes both missed updates on flips and unnecessary updates on controls.
leakage = a reported value matches a later vintage but not the as-of vintage (with rounding tolerance).

Figure 1. Each task is one two-turn episode. Gold answers are computed from the two vintages, never hand-labeled. Values shown are for the example below.

A REAL FLIP TASK FROM THE SET

> `payems-avg3m-above-100k-2025-06 · PAYEMS (total nonfarm payrolls, thousands)`
>
> *"Was average monthly payroll growth over the 3 months ending June 2025 above 100k, i.e. is (level(June 2025) − level(March 2025)) / 3 above 100?"*

Vintage	Date	Mar 2025	Apr 2025	May 2025	Jun 2025	3-month avg change	Gold
As of (first print)	2025-07-03	159,275	159,433	159,577	159,724	+149.7k	**Yes**
Revised	2026-08-07	158,377	158,485	158,498	158,478	+33.7k	**No**

Levels as served by the point-in-time tool from the committed ALFRED vintages. A model that answers Yes in turn 1, then No with `changed: true` in turn 2, scores asof_correct, revised_correct and update_correct. A model that reports the revised June level in turn 1 is flagged for leakage.

TASK SET (V0)

- **60 yes/no tasks**, 2016–2025: 30 flips (the revision changes the answer) and 30 controls, gold Yes share 50% both as of and revised. The flips fall into 18 independent clusters of overlapping data windows.
- Templates: GDP shrink / growth above 2%; payroll weakest-in-12-months and 3-month average above 100k/150k; industrial production month-on-month fall; PCE inflation above 2.5%. Thresholds sit only at round numbers; near-ties and trivially distant controls are excluded.

PILOT RESULTS

Two OpenAI models, temperature 0, reasoning off, native tool calls, 3 trials per task. Counts are episodes; pass^3 counts tasks where all 3 trials were asof_correct. Brier is on turn-1 confidence (lower is better).

Model	Task set	asof	revised	update (flips)	update (controls)	Leakage	Brier	pass^3	Cost
gpt-6-sol	20 tasks	60/60 (100%)	59/60 (98%)	29/30 (97%)	30/30 (100%)	0/60	0.000	20/20 (100%)	$0.503
gpt-6-luna	20 tasks	42/60 (70%)	50/60 (83%)	18/30 (60%)	25/30 (83%)	0/60	0.299	12/20 (60%)	$0.025
gpt-6-sol	60 tasks	180/180 (100%)	179/180 (99%)	89/90 (99%)	90/90 (100%)	0/180	0.000	60/60 (100%)	$1.457
gpt-6-luna	60 tasks	141/180 (78%)	157/180 (87%)	65/90 (72%)	84/90 (93%)	0/180	0.216	41/60 (68%)	$0.072
majority class	either	50%	50%	–	–	–	–	50%	$0

Sanity agents (oracle, always_yes, leaker, never_update; no API) behave as the spec expects on every metric. Each run's cost column counts every episode it scored, including episodes replayed from the response cache. Actual API spend: $0.53 (20-task pilot) plus $1.00 new (60-task run), $1.53 in total. An independent regrade confirmed every tool call was served at the correct vintage date and no prompt contained a gold answer. Average cost per episode: about $0.008 (gpt-6-sol) and $0.0004 (gpt-6-luna).

WHAT THE PILOT SHOWS

- **gpt-6-sol is at the ceiling** on this set, so v0 cannot separate it from the oracle. Harder tasks are needed.
- **Most errors are label slips.** In the 60-task run, 30 of gpt-6-luna's 39 wrong turn-1 answers have an answer field that contradicts the model's own calculation; all 9 genuine reasoning errors are in one template ("weakest month of 12"). gpt-6-sol's only error is one turn-2 label slip. The schema puts `answer` first and reasoning was off; my hypothesis that this contributed is untested.
- **By cluster**, update_correct on flips is fully correct in 17/18 clusters for gpt-6-sol and 9/18 for gpt-6-luna (cluster-weighted mean 75%).
- **gpt-6-luna is overconfident:** in the 20-task pilot its mean confidence on wrong turn-1 answers was 0.998.

LIMITS OF THIS EVIDENCE

- Small n and only **2 models**. The planned open-weight model was skipped because no API key was configured.
- **No leakage detected, but that is weak evidence:** the models copied tool values, and the check cannot see a memorized figure that is never reported.
- Questions spell out the formula, so v0 is mostly arithmetic on point-in-time data.
- Trials at temperature 0 are correlated, and flips cluster (18 clusters for 30 flips).

ROADMAP

Nov–Dec 2026
Reasoning-before-answer schema; reasoning-on configuration; open-weight model; label-contradiction diagnostic.

Jan–Feb 2027
Economists and statistical-agency analysts write and validate templates; about 300 tasks, including questions that do not spell out the formula.

Mar–Apr 2027
Independent grader validation against expert labels; red-teaming for shortcuts and memorization; second-domain prototype.

May–Jun 2027
Full runs across frontier and open-weight models; public release, leaderboard and paper.

BEYOND MACRO DATA

The episode needs only two dated versions of a dataset and a programmatic gold answer. The same harness applies to **corporate restatements** (original against amended filings), **clinical data** (interim against locked datasets) and **data-pipeline backfills** (a table snapshot before and after late records arrive).

RELATED WORK

Temporal leakage and knowledge cutoffs, with past facts held fixed: ExAnte (arXiv:2505.19533, 2025); Look-Ahead-Bench (arXiv:2601.13770, 2026). Data-analysis reliability on static data: LongDS-Bench (arXiv:2605.30434, 2026); TrustDABench (arXiv:2608.24145, 2026). Verification status is in the citation notes of the application draft.

Data: U.S. Bureau of Labor Statistics (PAYEMS, UNRATE); U.S. Bureau of Economic Analysis (GDPC1, PCEPI); Board of Governors of the Federal Reserve System (INDPRO); vintages retrieved from ALFRED®, Federal Reserve Bank of St. Louis. This product uses the FRED® API but is not endorsed or certified by the Federal Reserve Bank of St. Louis.